FELIX SHAW-BELL

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PROFILE

- Citizenship: New Zealand
- Current residence: Auckland, New Zealand
- Age: 24 years

A high-achieving and motivated mathematician with aspirations of making a lasting contribution to both mathematics and the wider community. I focus my mathematical abilities on real-world modelling, particularly in problems involving uncertainty. My strong academic record is complemented by industry experience, allowing me to apply myself in both a rigorous and practical manner.

EXPERIENCE

University of Auckland — March 2026 - June 2026
Graduate Teaching Assistant (part-time) — *Auckland, New Zealand*

- Delivered tutorials to classes of 20 to 40 students for first-year mathematics courses.

Shift Finance — February 2025 - August 2025
Data Scientist (full-time) — *Sydney, Australia (Remote)*

- Developed a machine learning model to assist and automate credit risk assessments.
- Consolidated multiple data sources into SQL tables that became widely used within the company.

University of Auckland — November 2024 - February 2025
Mathematics Research Scholar (full-time) — *Auckland, New Zealand*

- Worked under Professor Simon Harris and Dr Jesse Goodman to explore branching processes.
- Funded by the University of Auckland Summer Research Scholarship.

Shift Finance — November 2023 - November 2024
Data Analyst Intern (full-time & part-time) — *Sydney, Australia (On-site & Remote)*

- Built and deployed a company-specific Python library providing a rigorous machine learning pipeline.
- Carried out extensive feature engineering work to enhance model performance from existing datasets.

EDUCATION

University of Auckland — July 2025 - July 2026
MSc, Applied Mathematics (awaiting results) — *Auckland, New Zealand*

- Key areas of study: stochastic modelling, tipping point systems, approximation methods.
- Masters dissertation: '*Eigenmode approximations of noisy tipping-point systems*', supervised by Professor Graham Donovan. The project applied eigenmode approximations of the Fokker-Planck equation to estimate the risk of tipping in tipping point systems.

University of Auckland — February 2021 - November 2024
BSc (Hons), Applied Mathematics, First Class — *Auckland, New Zealand*

- Key areas of study: mathematical modelling, numerical methods, machine learning.
- Honours dissertation: '*Stochastic modelling of soil nitrogen in intensive agriculture*', supervised by Dr Steve Taylor. The project developed a system of stochastic differential equations to model the important issue of nitrogen pollution into agricultural waterways.

SKILLS

- Technical: MATLAB, Python, SQL, LaTeX.
- Languages: English (native), Spanish (elementary).

CONFERENCES AND WORKSHOPS

New Zealand Mathematics and Statistics Postgraduate Conference November 2025
'Eigenmode approximations of noisy tipping-point systems' (presentation) *Auckland, New Zealand*

AWARDS AND HONOURS

- University of Auckland Research Masters Scholarship (2025, stipend and free tuition).
- University of Auckland Summer Research Scholarship (2024, stipend).
- University of Auckland Young Scholars Program (2020).
- NZQA Calculus Scholarship (2020).
- NZQA Physics Scholarship (2020).
- Rotary Science & Technology Forum Sponsorship (2019).